

Sprint 3 Review

For Sprint 3, we successfully connected the upload feature to firebase, letting images and data to be stored and accessed without any problems. We also implemented an automatic and manual logout functionality which enhanced our app's security and session management. After that, we developed a feature that lets users delete clothing items from their wardrobe, improving the app's usability and cleaning up storage that is pointlessly being used up. One of our team members updated the home screen and upload image menu to maintain a consistent design and layout with the login and registration pages created a more professional user experience.

We also managed to connect the background removing program with `upload_service.dart`. To make them work, we had to rewrite the background class so that the main method, `removeBackground`, takes an image file, sends it to `remove.bg` web service, and saves the background free image as a new PNG in a temporary local directory. When these two classes work together, the `upload_service` class calls the `backgroundRemovalService` to get a clean transparent background version of the image making sure only the processed image is saved and associated with clothing item. After connecting the two classes and testing them, we added an automatic logout system that would check if the user was logged in. If they were, the system would log them out and show them a screen with a login and registration option.

During the previous sprint, we focused on implementing the connection between the registration screen and the home screen, reflecting the user story that a newly registered user should be directed straight to the home screen instead of having to log in. We also integrated the navigation logic, tested the transitions, and made sure that the data from the registration form was correctly passed to the home screen. While working on this, we encountered several challenges.

This sprint, we also added Firebase as the backend for real authentication in `login.dart` and replaced the mock login logic with `FirebaseAuth`. We also connected the registration screen so that after a user signs up and their info is saved to Firestore, they're signed out and redirected to the login page to sign in.

We implemented secure session handling by adding automatic logout with an optional "Remember Me" setting using `Firebase Auth` and local storage. We also polished the registration UI for consistency with our theme, ensured all fields are wired to controllers, and verified validations for smooth sign-up. Although some of this work landed a bit late this sprint, the features are complete and we're aligning our commits to stay on schedule next sprint.

One of the challenges our team faced was trying to hide the API key from the background removing service. To address this, we tried creating a `.env` file to store the API key but this proved to be harder than we thought. We had to add the `.env` file to `gitignore` to stop it from being committed and exploited by hackers. Then, we added an asset dependency onto `pubspec.yaml`. Finally, we let main know there would be a `.env` file, but in the end, things didn't

seem to work out, and we had to revert everything. Instead, we entered the API key straight into the background removing class and as of now, are still looking for a way to hide it. Another issue we ran into was that we had to upgrade our firebase account because the free version only stored data and no images. Since our app only takes in images, one of our team members stepped up and graciously paid to have images from our app stored into firebase.

When we met up with the client, we were given a couple user stories. We suggested some features to add to our app. Some of them got shot down while others seemed to really appeal to our client. We were told not to setup a mockup and if our code doesn't work, we shouldn't demo it. He was also not fond of the idea of being suggested to donate a clothing item but instead encouraged adding that in the settings so that if the user wants to get suggestions on donations, they can turn on that setting. Hard coding is also not a good practice when demoing incomplete code. If the app doesn't work, it doesn't work. User stories were supposed to come from the client and not us. This helped us this sprint because it gave us a guide to follow and let us know what steps we should take next to complete our project. It also cleared some misconceptions we had about the user stories.